

# BelkaCTF #7 — Compromised Air Traffic Control Workstation

Case INC-2026-0624-0001 · Examiner: niel · 2026-06-24 21:50 UTC

## CLASSIFICATION

**CONFIDENTIAL —  
AVIATION SAFETY  
IMPLICATIONS**

## TARGET SYSTEM

**TSO-ATC-CT412 (Windows  
10 build 19041  
(19041.1.amd64fre.vb\_releas)**

## COMPROMISE DETECTED

**YES — 3 CRITICAL severity  
findings, active C2  
established**

## INVESTIGATION WINDOW

**2025-07-22 18:50 – 19:18  
UTC (28 minutes)**

On 2025-07-22, an air traffic control workstation (TSO-ATC-CT412, user 'award') was compromised within 3 minutes of user login. A phishing email from ktso.sec@mailforce.net delivered a trojan (epxplorer.exe) masquerading as the Windows shell process explorer.exe. The malware established command-and-control to an attacker-operated Telegram Bot API behind Cloudflare infrastructure within 19 minutes of execution. At the time of compromise, ATC\_RADAR.exe — an aviation radar processing application — was actively running, meaning unauthorized code coexisted with live air traffic operations. This constitutes a potential aviation safety incident with possible regulatory reporting obligations under ICAO Annex 17. Immediate network isolation of the affected workstation and escalation to the aviation safety officer are required.

Chronological reconstruction of events from memory forensics artifacts. Color-coded by event type. Hover cards for source details.

2025-07-22 18:50:46 UTC

SYSTEM

System boot

volatility3 windows.info

2025-07-22 18:50:50 UTC

USER

User 'award' logs in — explorer.exe (PID 5116) starts

volatility3 windows.pslist

2025-07-22 18:51:58 UTC

USER

Microsoft Edge establishes HTTPS connection (browsing activity begins)

volatility3 windows.netscan

2025-07-22 18:53:11 UTC

MALWARE

MALWARE EXECUTION: epexplorer.exe (PID 9920) spawned by explorer.exe

volatility3 windows.pslist + MemProcFS

F-niel-001

2025-07-22 19:12:33 UTC

C2

C2 ESTABLISHED: epexplorer.exe → 104.211.247:443 (Telegram Bot API)

volatility3 windows.netscan

F-niel-002

2025-07-22 19:15:05 UTC

PROCESS

ATC\_RADAR.exe active — malware coexisting with aviation operations software

volatility3 windows.pslist + MemProcFS

F-niel-004

2025-07-22 19:18:53 UTC

SYSTEM

Memory dump captured — evidence acquisition point

volatility3 windows.info

● Malware / Compromise   ● Network / C2   ● Process Activity   ● File System   ● System Event   ● User Activity



| ID         | TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                   | CONFIDENCE            | TOOL                | EVIDENCE                                | CROSS-VALIDATION |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------|-----------------------------------------|------------------|
| F-niel-001 | Process epplorer.exe (PID 9920) was spawned by explorer.exe (PID 5116) at 2025-07-22 18:53:11 UTC, just 2 minutes 21 seconds after user login. The binary was executed from C:\Users\award\Downloads\epplorer.exe.                                                                                                                                                                                                                      | CRITICAL<br>HIGH conf | T1036.005 T1204.002 | volatility3 2.7.0 +<br>MemProcFS 5.17.8 | EVID-001         |
| F-niel-002 | epplorer.exe established an outbound HTTPS connection to 104.211.247:443 at 2025-07-22 19:12:33 UTC — 19 minutes after execution. The destination IP is a Cloudflare edge node fronting the attacker-controlled domain api-telegram-org.ctf.do. The malware uses Telegram Bot API (token: 8169267144:AAFwkhmXh71SMVcvFLantjTIwRbT9HdJdU) as its C2 channel, abusing legitimate cloud infrastructure to blend with normal HTTPS traffic. | CRITICAL<br>HIGH conf | T1102.002 T1573     | volatility3 2.7.0                       | EVID-001         |
| F-niel-003 | epplorer.exe was discovered at three virtual addresses in the memory filesan, including a partial browser download artifact (epplorer.exe435568.crdwn                                                                                                                                                                                                                                                                                   | HIGH<br>HIGH conf     | T1566.001 T1204.002 | volatility3 2.7.0                       | EVID-001         |

| ID         | TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CONFIDENCE            | TOOL      | EVIDENCE                                | CROSS-VALIDATION |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------|-----------------------------------------|------------------|
|            | load). The .crdownload extension is specific to Chromium-based browsers (Microsoft Edge in this environment — confirmed by extensive Edge artifacts throughout the filescan). Per CTF intelligence, the phishing email originated from ktso.sec@mailforce.net.                                                                                                                                                                                                                                |                       |           |                                         |                  |
| F-niel-004 | The compromised host (TSO-ATC-CT412, user 'award') runs ATC_RADAR.exe — an air traffic control radar application with multiple active processes and network connections to 8.8.8.8 (Google DNS-over-HTTPS). The workstation name (ATC-CT412) and application context confirm this is not a standard enterprise endpoint — it is an aviation operations system. Multiple ATC_RADAR.exe instances ran concurrently with explorer.exe, meaning the malware coexisted with active ATC operations. | CRITICAL<br>HIGH conf | T1003.001 | volatility3 2.7.0 +<br>MemProcFS 5.17.8 | EVID-001         |



| TYPE      | INDICATOR                                                        | CONTEXT           | FINDING    |
|-----------|------------------------------------------------------------------|-------------------|------------|
| SHA-256   | 3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35 | Evidence archive  | EVID-001   |
| SHA-256   | 7b45e797acabaf592784bf7efceb979e3f036d5b9dd47f52280c17e38864c2dd | Memory dump       | EVID-001   |
| MD5       | 75ea94b54420c39dcd3d8ce574ba9d34                                 | epxlorer.exe hash | F-niel-001 |
| Filename  | epxlorer.exe                                                     | Malware binary    | F-niel-001 |
| File Path | C:\Users\award\Downloads\epxlorer.exe                            | Execution path    | F-niel-001 |
| IP        | 104.21.1.247                                                     | C2 (Cloudflare)   | F-niel-002 |
| Domain    | api-telegram-org.ctf.do                                          | C2 domain         | F-niel-002 |
| Email     | ktso.sec@mailforce.net                                           | Phishing sender   | F-niel-003 |
| IP        | 172.17.4.12                                                      | Compromised host  | F-niel-004 |



#### EVID-001 – BelkaCTF\_7\_CASE250722\_KTSAERO.zip

SHA-256: 3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35

Source: [https://dl.ctf.do/BelkaCTF\\_7\\_CASE250722\\_KTSAERO.zip](https://dl.ctf.do/BelkaCTF_7_CASE250722_KTSAERO.zip)



| TOOL           | VERSION  | RUNTIME   | VALIDATED |
|----------------|----------|-----------|-----------|
| volatility3    | 2.7.0    | docker    | Validated |
| memprocs       | 5.17.8   | host_fuse | Validated |
| plaso          | 20240512 | docker    | Validated |
| mft-tools      | 1.2.0.0  | docker    | Validated |
| sift-vm        | ?        | ?         | Validated |
| session_canary | ?        | ?         | DEGRADED  |



| PROPERTY        | VALUE                                                     |
|-----------------|-----------------------------------------------------------|
| Hostname        | TS0-ATC-CT412                                             |
| Username        | award                                                     |
| Os              | Windows 10 build 19041 (19041.1.amd64fre.vb_release.1912) |
| Architecture    | x64                                                       |
| Processors      | 4                                                         |
| Host Ip         | 172.17.4.12                                               |
| Dump Time       | 2025-07-22 19:18:53 UTC                                   |
| Kernel Base     | 0xf80617e00000                                            |
| Nt Product Type | NtProductWinNt                                            |
| Source Tool     | volatility3 2.7.0 windows.info.Info                       |

Source: volatility3 2.7.0 windows.info.Info

Immediate containment and eradication steps per finding. Prioritize CRITICAL severity items first.

**CRITICAL** F-niel-001: Malware Execution: epexplorer.exe Masquerading as explorer.exe

Immediately isolate TSO-ATC-CT412 from all networks. Kill PID 9920. Quarantine C:\Users\award\Downloads\epexplorer.exe. Do NOT reboot before capturing full disk image — volatile evidence in memory will be lost.

**CRITICAL** F-niel-002: Command and Control via Fraudulent Telegram Bot API

Block outbound HTTPS to api-telegram-org.ctf.do at network perimeter. Block Telegram Bot API endpoints if not business-justified. Deploy SSL inspection for traffic to unknown Cloudflare IPs. Check firewall logs for additional connections from 172.17.4.12 in the 18:50-19:18 UTC window.

**HIGH** F-niel-003: Phishing Delivery Vector: Browser Download of Malicious Executable

Search mail server for emails from ktso.sec@mailforce.net — identify all recipients. Purge the phishing email from all inboxes. Block mailforce.net at email gateway. Initiate user awareness training for 'award' and ATC personnel. Review Edge browser security policies — consider disabling automatic execution of downloaded files.

**CRITICAL** F-niel-004: Critical Asset Compromised: Air Traffic Control Workstation

Escalate to aviation safety officer and CISO immediately. This may require mandatory reporting to the national aviation authority. Ground the workstation — do not allow it to process live radar data until forensic imaging is complete and the system is rebuilt from trusted media. Engage the ATC software vendor (ATC\_RADAR.exe) to verify software integrity. Review separation requirements between administrative workstations and operational ATC systems.





---

The following areas were not examined and represent uncertainty in the assessment. Each gap should be addressed before finalizing conclusions.

**GAP – Disk Image Not Analyzed**

Without filesystem forensics, persistence mechanisms, additional dropped files, and full file-system timeline cannot be confirmed.

**GAP – Network Capture Unavailable**

Encrypted C2 traffic cannot be analyzed for data exfiltration. Lateral movement cannot be assessed.

**GAP – Registry Hives Not Examined**

User activity artifacts and malware persistence locations (Run keys, scheduled tasks, services) remain unexamined.

**GAP – Malware Not Reverse-Engineered**

Full capability assessment of epexplorer.exe (keylogging, screen capture, lateral movement, exfiltration) is pending.

**GAP – Supplementary Evidence Not Analyzed**

The 5.6 GB supplementary file (SHA-256: e42f5254...) containing Android artifacts was not analyzed.

**GAP – Single Host Analysis Only**

Compromise of additional systems on the network cannot be ruled out without broader endpoint telemetry.

---

Examiner: niel · Case INC-2026-0624-0001

Hermes Forensics Agent · Report generated 2026-06-24 21:50 UTC

This report is auto-generated from structured case data. All findings require human review before finalization.